数理方程

第16周
20260612

格林函数(续)
微扰论
非线性PDE

波动方程格林函数



一维 $u_{tt}-a^2u_{xx}=f(x,t)$, $0<x<l$, $t>0$

自行阅读 $u|_{x=0}=\mu(t)$, $u|_{x=l}=\nu(t)$, $t>0$

$u|_{t=0}=\phi(x)$, $\left.\frac{\partial u}{\partial t}\right|_{t=0}=\psi(x)$, $0<x<l$

$$\Rightarrow \frac{\partial^2 G(x,t;x',t')}{\partial t^2}-a^2\frac{\partial^2 G}{\partial x^2}=\delta(x-x')\delta(t-t'),\quad 0<x,x'<l,\ t,t'>0$$

外力作用在t'时刻

$$G|_{x=0}=G|_{x=l}=0,\quad G|_{t=0}=\left.\frac{\partial G}{\partial t}\right|_{t=0}=0$$



$$\Rightarrow G(x,t;x',t')|_{t<t'}=0,$$

$$G|_{t\geq t'}=?$$

如果已知 $G\Rightarrow u$.

格林公式: $L=\frac{\partial^2}{\partial t^2}-a^2\frac{\partial^2}{\partial x^2}$

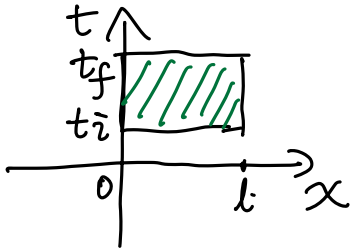
$$uL(v)-vL(u)=uv_{tt}-vu_{tt}-a^2(uv_{xx}-vu_{xx})$$

$$=(uv_t-vu_t)_t-a^2(uv_x-vu_x)_x$$

$$\Rightarrow \int_{t_i}^{t_f}\int_0^l [uL(v)-vL(u)]\,dx\,dt$$

$$= \int_0^l (uv_t - vu_t)\,dx\Big|_{t_i}^{t_f} - a^2\int_{t_i}^{t_f}(uv_x - vu_x)\Big|_0^l dt$$

时间边界　　空间边界



$u \to u,\quad v \to G$

$$\Rightarrow u(x,t) = \int_0^t dt' \int_0^l dx'\, G(x,t;x',t') f(x',t')\,dt$$

$$+\int_0^l dx' \left[G(x,t;x',0)\psi(x') - \phi(x')\frac{\partial G(x,t;x',t')}{\partial t'}\Big|_{t'=0}\right]$$

$$-a^2\int_0^t dt' \left[v(t)\frac{\partial G(x,t,x',t')}{\partial x'}\Big|_{x'=l} - \mu(t)\frac{\partial G}{\partial x'}\Big|_{x'=0}\right]$$

无界空间 $-\infty < x < \infty$

$$u(x,t) = \int_0^t dt' \int_{-\infty}^{+\infty} dx'\, G(x,t;x',t') f(x',t')$$

$$+\int_{-\infty}^{+\infty} dx' \left[G(x,t;x',0)\psi(x') - \phi(x')\frac{\partial G(x,t;x',t')}{\partial t'}\Big|_{t'=0}\right]$$

$$\begin{cases} \dfrac{\partial^2 G(x,t;x',t')}{\partial t^2} - a^2\dfrac{\partial^2 G}{\partial x^2} = \delta(x-x')\delta(t-t'), & t,t' \geq 0 \\ G|_{t=0} = \dfrac{\partial G}{\partial t}\Big|_{t=0} \\ G|_{t\leq t'} = 0 \end{cases}$$

$t > t'$

$$\begin{cases} G_{tt} - a^2 G_{xx} = 0 \\ G|_{t=t'} = 0, \quad \dfrac{\partial G}{\partial t}\Big|_{t=t'} = \delta(x-x') \end{cases}$$

$\delta(x-x')\delta(t-t')$

$$\frac{\partial G}{\partial t}\Big|_{t=t'+} - \frac{\partial G}{\partial t}\Big|_{t=t'-} = \int_{t'-}^{t'+}\frac{\partial^2 G}{\partial t^2}\,dt$$

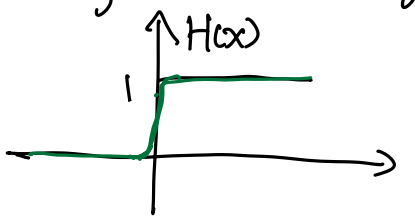
$=0$　　$= \delta(x-x')$

$t'=0,\ x'=0$

$G_1 = f(x-at)+g(x+at) \xrightarrow{G|_{t=0}=0} f(x-at)-f(x+at)$

$\delta(x) = \frac{\partial G}{\partial t}\Big|_{t=0} = -af'(x)-af'(x) = -2af'(x)$

$\Rightarrow f'(x) = -\frac{1}{2a}\delta(x)$

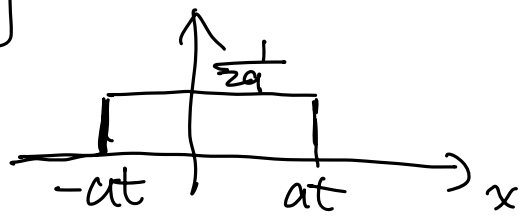


$\int_a^b \frac{dH}{dx}dx = H(b)-H(a)$

$f(x) = -\frac{1}{2a}H(x)+C$

$G(x,t;0,0) = \left[-\frac{1}{2a}H(x-at)+C\right]-\left[-\frac{1}{2a}H(x+at)+C\right]$

$= \frac{1}{2a}\left[H(x+at)-H(x-at)\right]$



$G(x,t;x',t') = \frac{1}{2a}\left[H(x-x'+a(t-t'))\right.$
$\left.-H(x-x'-a(t-t'))\right]$

$\frac{\partial G}{\partial t}\Big|_{t'=0} = \frac{1}{2a}\left[-aH'(x-x'+at)-aH'(x-x'-at)\right]$

$= -\frac{1}{2}\left[\delta(x-x'+at)+\delta(x-x'-at)\right]$

$f=0$

$\Rightarrow u(x,t) = \int_{-\infty}^{+\infty} dx'\left[G(x,t;x',0)\psi(x')-\phi(x')\frac{\partial G(x,t;x',t')}{\partial t'}\Big|_{t'=0}\right]$

$= \frac{1}{2a}\int_{x-at}^{x+at}\psi(x')dx' + \frac{1}{2}\left[\phi(x+at)+\phi(x-at)\right]$

三维无界空间 Green's function.

$u_{tt}-a^2\nabla^2 u(\vec{r},t) = f(\vec{r},t),\ t>0$

$\begin{cases} u|_{t=0} = \phi(\vec{r}), & \frac{\partial u}{\partial t}\Big|_{t=0} = \psi(\vec{r}) \end{cases}$

↑也可用 Fourier 变换解.

$$\frac{\partial^2 G(\vec{r},t;\vec{r}',t')}{\partial t^2} - a^2\nabla^2 G = \delta(\vec{r}-\vec{r}')\delta(t-t'),\quad t,t'\geq 0$$

$$\begin{cases} G|_{t=0} = \dfrac{\partial G}{\partial t}\Big|_{t=0} = 0 \end{cases}$$

$$\Rightarrow G|_{t\leq t'} = 0$$

$$\begin{cases} \dfrac{\partial^2 G}{\partial t^2} - a^2\nabla^2 G = 0, \quad t > t' \\ G|_{t=t'} = 0,\ \dfrac{\partial G}{\partial t}\Big|_{t=t'} = \delta(\vec{r}-\vec{r}') \end{cases}$$

$\vec{r}'=0,\ t'=0 \Rightarrow G(r,t;0,0)$

$$G|_{t=0} = 0,\quad \frac{\partial G}{\partial t}\Big|_{t=0} = \delta(\vec{r})$$

球对称: $G = G(r,t)$

$$\frac{\partial^2 G}{\partial t^2} - a^2\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial G}{\partial r}\right) = 0, \qquad G(r,t) = \frac{h(r,t)}{r}$$

$$= \frac{1}{r}\left(\frac{\partial^2 h}{\partial t^2} - a^2\frac{\partial^2 h}{\partial r^2}\right) = 0$$

$$G(r,t) = \frac{f(r-at)+g(r+at)}{r} \xrightarrow{G|_{t=0}} \frac{1}{r}[f(r-at) - f(r+at)]$$

$$\frac{\partial G}{\partial t}\Big|_{t=0} = \frac{1}{r}[-af'(r) - af'(r)] = -\frac{2a}{r}f'(r) = \delta(r)$$

$r\neq 0$: $f'(r)=0$, $f(r) = \text{const}$. 取 $f(r)=0$

$$1 = \iiint_{r<R}\delta(\vec{r})\,d\vec{r} = -\iiint_{r<R}\frac{2a}{r}f'(r)\,4\pi r^2\,dr = -8\pi a\int_0^R f'(r)r\,dr$$

$\int_0^R r\,df$

$$= 8\pi a\left[-f(r)\,r\Big|_0^R + \int_0^R f(r)\,dr\right]$$

$f(R)=0$

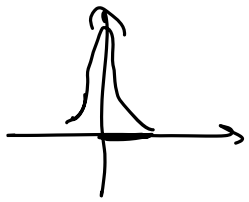
$= 8\pi a \int_0^R f(r)\,dr$

$= 8\pi a \frac{C}{2}$

$= 4\pi a C$

$\Rightarrow C = \frac{1}{4\pi a}$

$f(r) = C\delta(r)$



$\Rightarrow G(\vec{r};t;0,0) = \frac{1}{r}[f(r-at) - f(r+at)]$

$= \frac{1}{4\pi a r}[\delta(r-at) - \delta(r+at)]$

$= \frac{1}{4\pi a r}\delta(r-at)$

$G(\vec{r};t;\vec{r}',t') = \frac{1}{4\pi a|\vec{r}-\vec{r}'|}\delta(|\vec{r}-\vec{r}'| - a(t-t'))$

三维波动方程 格林函数

$\delta(at') = \frac{\delta(t')}{|a|}$

$$u(\vec{r},t) = \int_0^t \iiint_{-\infty}^{+\infty} G(\vec{r},t;\vec{r}',t') f(\vec{r}',t')\,d\vec{r}'dt'$$

$$+ \iiint_{-\infty}^{+\infty} \left[G(\vec{r},t;\vec{r}',0)\psi(\vec{r}')\,d\vec{r}' - \phi(\vec{r}) \frac{\partial G(\vec{r},t;\vec{r}',t')}{\partial t'}\Big|_{t'=0} \right] d\vec{r}'$$

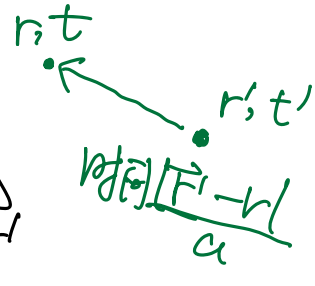
$f=0$: $u(\vec{r},t) = \frac{1}{4\pi a}\iint_{|\vec{r}'-r|=at} \frac{\psi(\vec{r}')}{|\vec{r}-\vec{r}'|}\,ds' + \frac{\partial}{\partial t}\left[\frac{1}{4\pi a}\iint_{|\vec{r}'-r|=at} \frac{\phi(\vec{r}')}{|\vec{r}-\vec{r}'|}\,ds'\right]$

泊松公式

球面面积元

$\phi = \psi = 0,\ f \neq 0$

$\Rightarrow u(\vec{r},t) = \frac{1}{4\pi a^2}\iiint_{|\vec{r}'-\vec{r}|\le at} \frac{f(\vec{r}', t - \frac{|\vec{r}'-r|}{a})}{|\vec{r}'-\vec{r}|}\,d\vec{r}'$



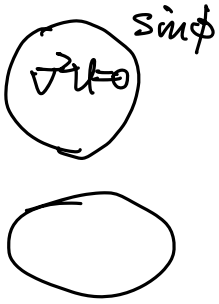
推迟势

电磁波

二维：降维法

微扰法 perturbation theory.

对理想情况的偏离较小时.



$$\begin{cases} \nabla^2 u = 0, & r < R + \epsilon\cos\phi \\ u\big|_{r=R+\epsilon\cos\phi} = \sin\phi \end{cases}$$

ϵ 小量

$\nabla^2 u^{(0)} = \nabla^2 u^{(1)} = \cdots = 0$

$$u(r,\phi) = u^{(0)}(r,\phi) + \epsilon u^{(1)}(r,\phi) + \epsilon^2 u^{(2)} + \epsilon^3 u^{(3)} + O(\epsilon^4)$$

$\epsilon = 0.1$, $\epsilon^4 \sim 10^{-4}$

$$u\big|_{r=R+\epsilon\cos\phi} = u^{(0)}\big|_R + \epsilon u^{(1)}\big|_R + \epsilon^2 u^{(2)}\big|_R + \epsilon^3 u^{(3)}\big|_R$$

$u|_R$

$$+ \left[\frac{\partial u^{(0)}}{\partial r} + \epsilon\frac{\partial u^{(1)}}{\partial r} + \epsilon^2\frac{\partial u^{(2)}}{\partial r} + \cdots\right]\epsilon\cos\phi$$

$\frac{\partial u}{\partial r}\big|_R \epsilon\cos\phi$

$$+ \left[\frac{1}{2}\frac{\partial^2 u^{(0)}}{\partial r^2} + \frac{1}{2}\epsilon\frac{\partial^2 u^{(1)}}{\partial r^2} + \cdots\right](\epsilon\cos\phi)^2$$

$\frac{1}{2}\frac{\partial^2 u}{\partial r^2}\big|_R (\epsilon\cos\phi)^2$

$$+ \left[\frac{1}{3!}\frac{\partial^3 u^{(0)}}{\partial r^3} + \cdots\right](\epsilon\cos\phi)^3 + \cdots$$

$$= \sin\phi$$

ϵ^0 级：$u^{(0)}\big|_{r=R} = \sin\phi \Rightarrow u^{(0)} = \frac{r}{R}\sin\phi$

ϵ^1 级：$u^{(1)}\big|_R + \frac{\partial u^{(0)}}{\partial r}\Big|_R \cos\phi = 0 \Rightarrow u^{(1)}\big|_R = -\frac{\sin\phi}{R}\cos\phi$

$$= -\frac{\sin 2\phi}{2R}$$

$$\Rightarrow u^{(1)} = -\frac{r^2}{2R^3}\sin 2\phi$$

ϵ^2 阶：$u^{(2)}\big|_R + \underbrace{\frac{\partial u^{(1)}}{\partial r}\Big|_R}_{\text{已知}} \cos\phi + \underbrace{\frac{1}{2}\frac{\partial u^{(0)}}{\partial r^2}\Big|_R}_{\text{已知}=0} \cos^2\phi = 0$ "递推关系"

$$\Rightarrow u^{(2)}\big|_R = -\frac{\partial u^{(1)}}{\partial r}\Big|_R \cos\phi = \frac{1}{R^2}\sin 2\phi\cos\phi = \frac{1}{2R^2}(\sin 3\phi + \sin\phi)$$

$$\Rightarrow u^{(2)}(r,\phi) = \frac{r^3}{2R^5}\sin 3\phi + \frac{r}{2R^3}\sin\phi$$

同理 $u^{(3)} = -\frac{5r^4}{8R^7}\sin 4\phi - \frac{3r^2}{4R^5}\sin 2\phi$

$$\Rightarrow u(r,\phi) = \frac{r}{R}\sin\phi - \epsilon\frac{r^2}{2R^3}\sin 2\phi + \epsilon^2\left(\frac{r^3}{2R^5}\sin 3\phi + \frac{r}{2R^3}\sin\phi\right)$$
$$+ \epsilon^3\left(-\frac{5r^4}{8R^7}\sin 4\phi - \frac{3r^2}{4R^5}\sin 2\phi\right) + \cdots$$

地球外引力场计算类似

$f(\theta) \propto P_2(\cos\theta)$ 与地球自转离心力有关

地球形状 $R(\theta) = R_0\left(1 + \epsilon f(\theta)\right)$

$\epsilon = \frac{a-b}{a}$ ← 极半径 / ↑ 赤道半径

引力势 $u(r,\theta) = -\frac{GM}{r}\left[1 - \frac{2}{3}\epsilon\left(\frac{R_0}{r}\right)^2 P_2(\cos\theta) + O(\epsilon^2)\right]$

$\approx 1.0826 \times 10^{-3}$

微扰法用于(弱)非线性PDE

例: $\nabla^2 u + \epsilon u^2 = 0$, $u|_{r=R} = \cos\phi$



$\nabla^2 u = -\epsilon u^2$

$u = u^{(0)} + \epsilon u^{(1)} + O(\epsilon^2)$

$\nabla^2(u^{(0)} + \epsilon u^{(1)} + \cdots) + \epsilon(u^{(0)} + \epsilon u^{(1)})^2 = 0$

$\Rightarrow$ ϵ^0项: $\nabla^2 u^{(0)} = 0$

ϵ^1项: $\nabla^2 u^{(1)} + (u^{(0)})^2 = 0$

BC: $u^{(0)}|_R = \cos\phi$, $u^{(1)}|_R = 0$

$\Rightarrow u^{(0)} = \frac{r}{R}\cos\phi$

$\nabla^2 u^{(1)} = -(u^{(0)})^2 = -\frac{r^2}{R^2}\cos^2\phi = -\frac{r^2}{2R^2}(1+\cos 2\phi)$

$\Rightarrow u^{(1)} = \frac{R^2}{32}(1-\frac{r^4}{R^4}) + \frac{r^2}{24}(1-\frac{r^2}{R^2})\cos 2\phi$

同理求 $u^{(2)}, u^{(3)} \cdots$

若 $u^{(0)} \to A u^{(0)}$ 齐次A

则 $u^{(1)} \to A^2 u^{(1)}$

$u^{(2)} \to A^3 u^{(2)}$

$\vdots$

$u^{(n)} \to A^{n+1} u^{(n)}$

$\Downarrow$

$A[u^{(0)} + A\epsilon u^{(1)} + A^2\epsilon^2 u^{(2)} + \cdots + A^n \epsilon^n u^{(n)} + \cdots]$

非线性PDE

非线性介质

非线性扩散方程 $u_t=(D(u)u_x)_x$

流体 Navier-Stokes $u_t=(u^n u_x)_x$

多孔介质方程

(地下水位方程)

线性

$L(u)=0 \Rightarrow L(cu)=0$

$L(u_1)=L(u_2)=0 \Rightarrow L(c_1u_1+c_2u_2)=0$

唯一性，单值性

解析解困难，一般理论和系统解法缺乏

湍流，孤波，孤立子

冲击波(激波)

光滑初始条件能演化出奇异性

↑ 非线性PDE典型现象之一

激波及其速度的简单模型

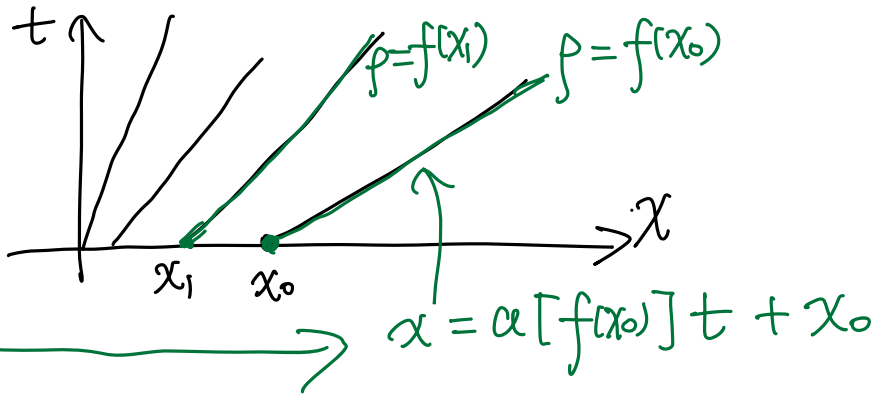
密度波方程

$$\begin{cases} \dfrac{\partial \rho}{\partial t} + a(\rho)\dfrac{\partial \rho}{\partial x} = 0 \\ \rho|_{t=0} = f(x) \end{cases}$$

$a(\rho) = \rho$

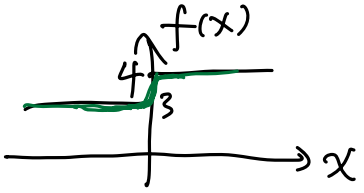
ρ是解，$C\rho$不是解

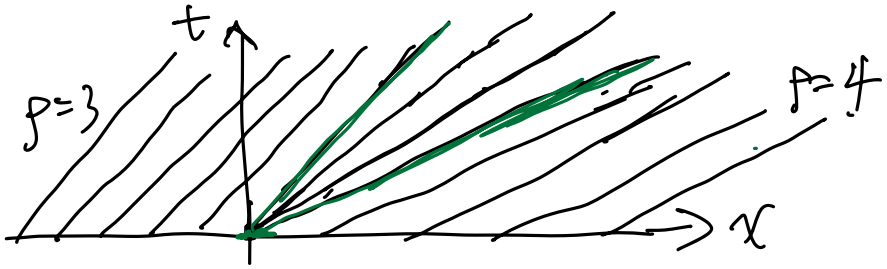
特征线 $\dfrac{dx}{dt} = a(\rho)$

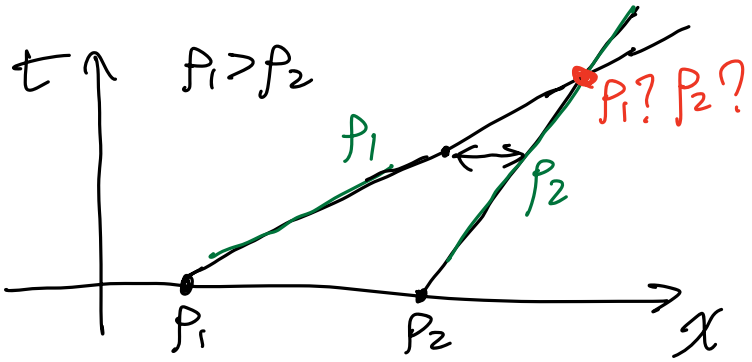


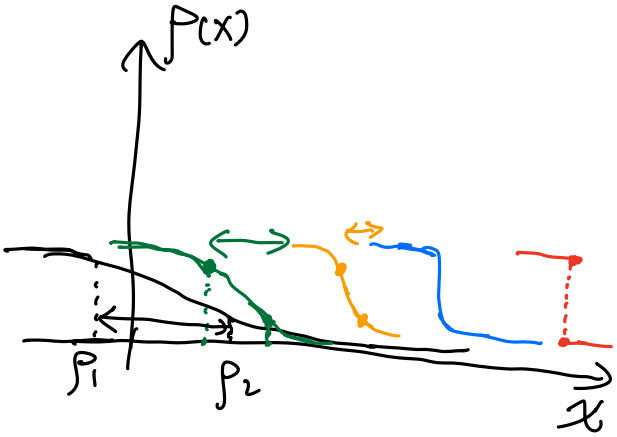
沿特征线 $\dfrac{d\rho}{dt} = \rho_t + \dfrac{dx}{dt}\rho_x = \rho_t + a(\rho)\rho_x = 0$

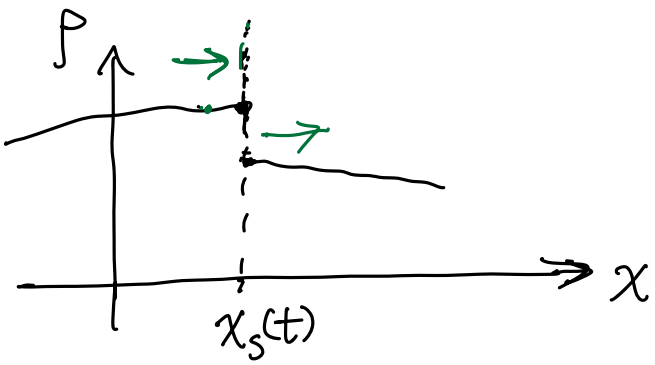
例：
$$\begin{cases} \rho_t + \rho \cdot \rho_x = 0 \\ \rho|_{t=0} = f(x) = \begin{cases} 3 & x < 0 \\ 4 & x > 0 \end{cases} \end{cases}$$











$$p(x_s-,t)\left[u(x_s-)-\frac{dx_s}{dt}\right]=p(x_s+,t)\left[u(x_s+)-\frac{dx_s}{dt}\right]$$

$$\Rightarrow \frac{dx_s}{dt}=\frac{p(x_s+,t)u(x_s+)-p(x_s-)u(x_s-)}{p(x_s+)-p(x_s-)}$$

$$=\frac{q(x_s+)-q(x_s-)}{p(x_s+)-p(x_s-)}=\frac{[q]}{[p]}$$

← 记为

激波速度 Rankine-Hugoniot formula

真实的气体中的激波速度公式与此极为相似.

线性的世界简单，可预测，也略枯燥。

非线性的世界复杂，难解，也更丰富，有更多可能！

例如从光滑初始条件演化出奇异性。